MIL-B-43407
24 January 1966

MILITARY SPECIFICATION

BARS: NONSPARKING AND NONMAGNETIC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers three types of bars made of nonferrous alloys which have low sparking hazard and low magnetic permeability.

1.2 Classification.- Bars shall be of the following types and sizes as specified (see 6.2).

Type I - Crowbar - 60 inches long
Type II - Pinchbar - 18 inches long by 3/4 inch diameter
Type III - Wrecking bar -
Size 1 - 18 inches by 5/8 inch diameter
Size 2 - 24 inches by 3/4 inch diameter
Size 3 - 36 inches by 7/8 inch diameter

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-T-580 - Tool Steel, Carbon and Carbon-Vanadium

MILITARY

MIL-G-5572 - Gasoline Aviation; Grades 80/87, 91/97, 100/130, 115/145

MIL-H-15424 - Hand Tools, Packaging Of

MIL-I-17214 - Indicator, Permeability, Low-Mu (Go-No-Go)

FSC 5120

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:21Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

Fed. Std. No. 151 - Metals, Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified (see 6.2), before production is commenced, a sample bar covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.1.1 Nonsparking preproduction sample.- Unless otherwise specified (see 6.2), before production is commenced and within 30 days after the date of award of the contract or order, a test specimen of the material (approximately 1/4 inch in diameter by 2 inches long) to be used in the manufacture of the items covered by this specification shall be submitted to the Government for the nonsparking test as specified in 4.4.3 (see 6.3).

3.2 Material.- The nonsparking and nonmagnetic bars shall be forged or cast from nonferrous alloy that will meet the requirements of this specification.

3.2.1 Magnetic permeability.- The bar material shall have a magnetic permeability not exceeding L2 of the indicator scale when tested in accordance with 4.4.2.

3.2.2 Nonsparking.- The nonsparking bar material shall not cause an explosion when tested in accordance with 4.4.3.

3.3 Design.- The bars shall be in accordance with the design and dimensions shown on figures 1, 2 and 3 and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:21Z
Check the source to verify that this is the current version before use.

3.3.1 Type I - crowbar.- The crowbar shall have a round handle tapering slightly toward the handle end. Not less than 6 inches of the working end of the crowbar shall have a hardness of Rockwell C32-40 when tested as specified in 4.4.1. Sides of the beveled edge shall be perpendicular to sides on the wedge point end (see figure 1). The crowbar shall not fracture or take a permanent set when tested as specified in 4.4.4.

3.3.2 Type II - pinchbar.- The pinchbar shall have one end round and tapered for bringing holes into alignment and the other end shall be flattened and offset (see figure 2). The work ends of the bar shall have a hardness of Rockwell C32-40 when tested as specified in 4.4.5. The ends shall not fracture or take a permanent set when tested as specified in 4.4.5.

3.3.3 Type III - wrecking bar.- The wrecking bars shall have one end bent in the form of a gooseneck and the opposite end in the form of a pinch point. The gooseneck shall be provided with a slotted claw suitable for pulling nails. The opposite end shall be wedge shaped and shall be offset at an angle of 30 degrees plus or minus 5 degrees from the longitudinal axis of the bar (see figure 3). The wrecking bars shall have a hardness of Rockwell C28-40 for a minimum distance of 2-3/4 inches from each end when tested as specified in 4.4.1. The bars shall not fracture or take a permanent set when tested as specified in 4.4.6.

3.3.3.1 Dimensions.- The wrecking bars shall be of the sizes and within the tolerances shown on figures 1, 2 and 3.

3.4 Finish.- The striking ends and the cutting edges shall be smoothly machined or ground. Forged, wrought or cast surfaces shall be free from fins, scale, cracks, seams or burrs.

3.5 Marking.- Each bar shall be marked on the handle in a plain and permanent manner with the name, abbreviation or chemical symbol of the alloy used in the fabrication of the bars. Each bar shall also be marked with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.6 Workmanship.- The completely fabricated bars shall be free of burrs, cracks, pits, slivers and gouges. Flash lines shall be removed sufficiently to blend smoothly with adjacent surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:21Z
Check the source to verify that this is the current version before use.

4.2 Preproduction sample inspection.- When a preproduction sample is required it shall be examined for the defects in table I, dimensions specified, and tested in accordance with 4.4.1, 4.4.2, and 4.4.4, 4.4.5 or 4.4.6, as applicable.

4.3 Sampling for inspection.- Sampling for inspection shall be made in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated hereinafter.

4.3.1 End item inspection.- The inspection lot shall consist of all of the bars of the same type and size offered for inspection at one time. The sample unit shall be one bar.

4.3.1.1 Visual examination.- Examination shall be made of bars for the defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect
Finish	Not as specified Not free of burrs, cracks, pits, slivers or gouges
Construction	Not type specified Working edges damaged
Marking	Missing, incomplete not legible

4.3.1.2 Dimensional examination.- Examination shall be made of bars to determine compliance with dimensions specified. Any dimension not within specified tolerances shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.1.3 End product testing.- When no preproduction sample is required, one bar of each type from the initial lot shall be tested in accordance with 4.4.2. Each lot shall be tested in accordance with 4.4.1 and with 4.4.4, 4.4.5 or 4.4.6 as applicable. The inspection level shall be S-2 with an AQL of 1.5 defects, expressed in terms of defects per hundred units. Failure to pass any tests shall be cause for rejection of the entire lot.

4.3.2 Examination of preparation for delivery.- An examination shall be made for preservation, packaging, packing and marking in accordance with MIL-H-15424. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 20261O-21T21:21Z
Check the source to verify that this is the current version before use.

4.4 Tests.-

4.4.1 Hardness tests.- Hardness tests shall be in accordance with Method 243.1 of Fed Std. No. 151 to determine compliance with 3.3.1, 3.3.2 and 3.3.3.

4.4.2 Magnetic permeability.- The bar material shall be tested for magnetic permeability with a permeability indicator as shown in MIL-I-17214 and in accordance with the operating instructions accompanying each indicator. The indicator reading shall comply with 3.2.1.

4.4.3 Sparking test.- (see 6.3).

4.4.3.1 Apparatus.- The sparking test shall be conducted in an explosion chamber having a volume of 4.5 cubic feet. The chamber shall be furnished with the following: a hot plate for vaporizing gasoline; a means for admitting gasoline and oxygen at atmospheric pressure; a fan for mixing the gasoline vapors and oxygen; two electrodes or other suitable means for providing an electric spark; a 3-inch diameter by 1/4 inch thick metal wheel having a coarsely knurled surface on the circumference and rotating at a speed of 10,000 revolutions per minute; and a fixture for holding the test specimen against the circumference of the metal wheel while applying pressure from the outside of the chamber. The metal wheel shall be made from carbon tool steel conforming to class WI-09 of QQ-T-580 and heat-treated to a hardness of Rockwell C60 to 65. The gasoline to be used during these tests shall be aviation gasoline conforming to grade 100/130 of MIL-G-5572. An oxygen analyzer shall be used for determining the percentage of oxygen in the chamber.

4.4.3.2 Procedure.- A test specimen approximately 1/4 inch in diameter by 2 inches long, shall be made from material used in the sample tool and subjected to the sparking test. With the specimen mounted in the holder fixture and the chamber sealed, oxygen shall be admitted into the chamber until the oxygen content in the air reaches 50 percent. A pan resting on the hot plate shall then be filled with 20 cubic centimeters of gasoline through an opening at the top of the chamber, the opening sealed, the fan turned on, and the gasoline evaporated. After the gasoline is completely evaporated, the fan shall be turned off and the specimen held against the rotating wheel with a force of 2.5 pounds. If no explosion occurs before 5 seconds, the mixture shall then be exploded by means of the electric spark to demonstrate the presence of an explosive mixture. Three runs shall be made with the test specimen, and if the specimen explodes the gasoline and oxygen-enriched air mixture during any of these runs, the tools represented by the specimen tested shall be disapproved for use as nonsparking tool.

4.4.4 Type I - crowbars.- The sample crowbars shall be subjected to the following tests: Two inches of the pointed end (or work end) shall be held securely or supported in a suitable holder, such as a vise, and a test load of 200 pounds applied at approximately the midpoint of the handle section in a direction perpendicular to the body, to determine compliance with 3.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:21Z
Check the source to verify that this is the current version before use.

4.4.5 Type II - pinch bars.- The sample pinch bars shall be subjected to the following tests to determine compliance with 3.3.2. The entire offset end of the bar (dimension "A" of figure 2) shall be rigidly secured by means of a vise or clamp and a 50 pound test load shall be applied to the body approximately 2 inches from the tapered end in a direction perpendicular to the body. The manner of applying the load shall be such as to straighten the offset. Three inches of the tapered end shall be clamped and the bar shall be subjected to the same test, except that an 18 pound load shall be applied.

4.4.6 Type III - wrecking bars.- One and one-half inches of the wedge shaped offset end of the type III bars shall be rigidly secured by means of a vise or clamp, and the applicable test load specified in figure 3 shall be applied to the body at a point 14 inches for size 1, 20 inches for size 2, 32 inches for size 3, from the clamped end, in a direction perpendicular to the body. The manner of applying the load shall be such as to straighten the offset. The claw end shall also be subjected to the same test. These tests are to determine compliance with 3.3.3.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B and C.- Preservation, packaging, packing and marking for each level of protection shall be in accordance with the applicable requirements of MIL-H-15424 (see 6.2).

6. NOTES

6.1 Intended use.- The various type bars are intended for use in hazardous areas for moving objects and for wrecking purposes.

6.2 Ordering data.- Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size (as applicable) of bars required (see 1.2.1).
(c) When preproduction sample is not required (see 3.1).
(d) When requirement for nonsparking preproduction sample is not required (see 3.1.1).
(e) Selection of the applicable levels of preservation, packaging and packing (see 5.1).

6.3 Nonsparking preproduction sample.- The contracting officer, at the time of award of the contract or order, shall contact the Bureau of Ships, Code 621, Washington, D. C. or Marine Engineering Laboratories, Code 280, Annapolis, Maryland 21204 and make arrangements for the performance of the test and obtaining the results of the test. The test specimen should be sent to Philadelphia Naval Shipyard, Code 305, Philadelphia, Pa. (see 3.1.1 and 4.4.3).

Source: https://assist.dla.mil -- Downloaded: 20261O-21T21:21Z
Check the source to verify that this is the current version before use.

Custodians:
Army - GL
Navy - SH
Air Force - 69

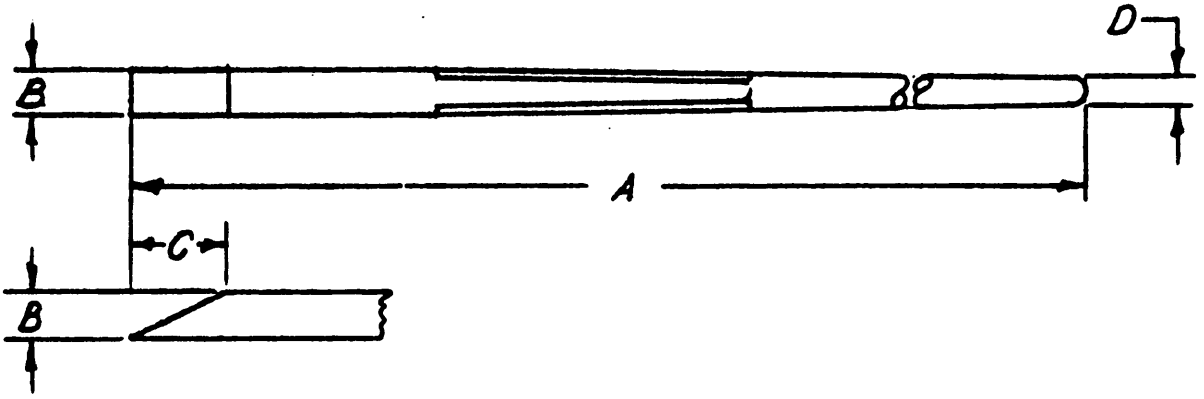
Review activities:
Army - WC
Navy - SH, WP
Air Force - 69

User activities:
Army - EL, MO, CE
Navy - YD, MC

Preparing activity:
Army - GL
Project No. 5120-0503

Review and user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Documents.

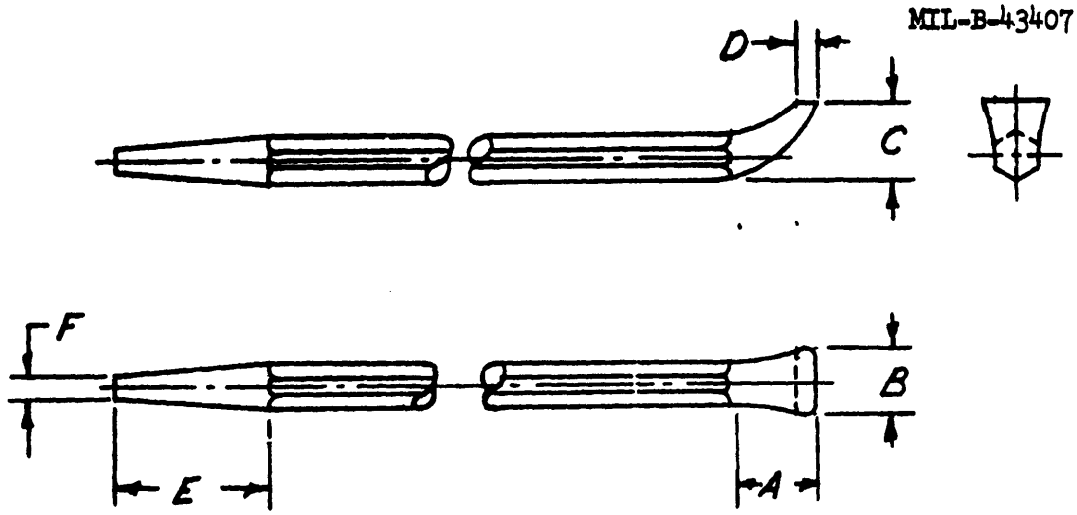
Source: https://assist.dla.mil -- Downloaded: 202610-21T21:21Z
Check the source to verify that this is the current version before use.



A		B	C	D	AREA OF HARDNESS FROM WEDGE-POINT	WEIGHT		TEST LOAD
MINIMUM	MAXIMUM	$\pm \frac{1}{16}$	MINIMUM	$\pm \frac{3}{16}$	MINIMUM	MINIMUM	MAXIMUM	MINIMUM
INCHES	INCHES	INCHES	INCHES	INCH	INCHES	POUNDS	POUNDS	POUNDS
60	62	$1\frac{1}{4}$	3	$\frac{9}{8}$	6	17	19	200

FIGURE 1 TYPE I CROWBAR, WEDGE POINT

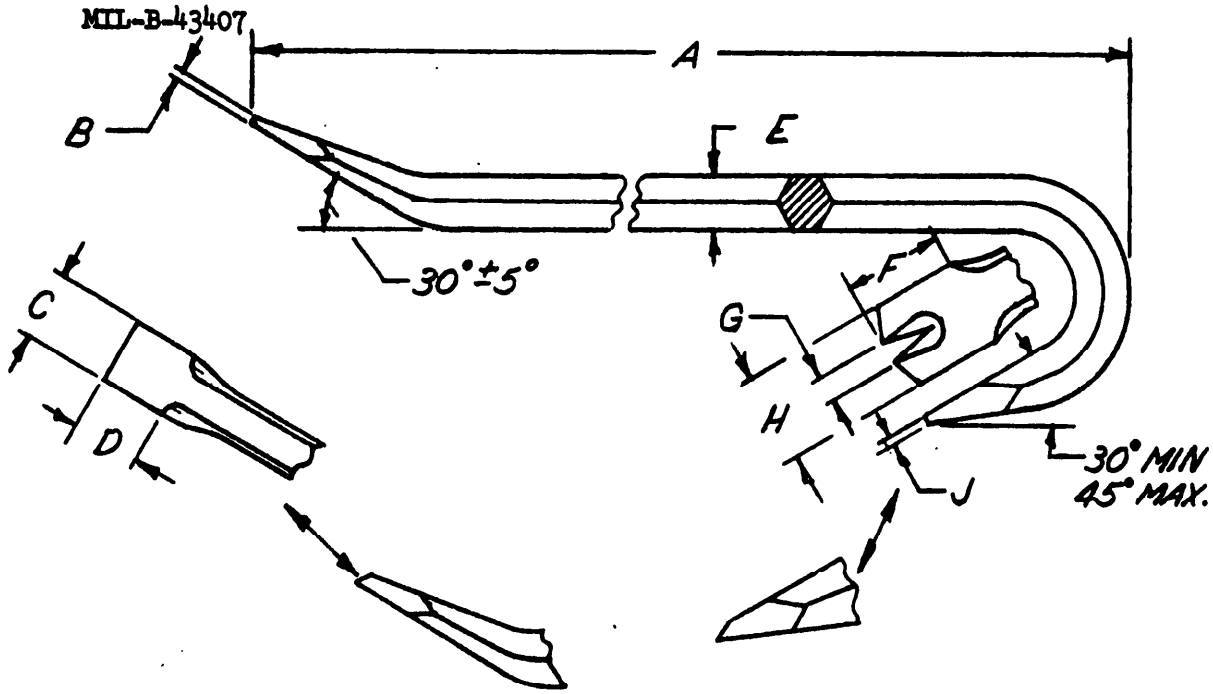
Source: https://assist.dla.mil -- Downloaded: 20261O-21T21:21Z
Check the source to verify that this is the currentversion before use.



DIMENSION BETWEEN OPPOSITE FLATS ±1/32	LENGTH OVERALL ±1/2	DIMENSIONS						TEST LOAD MINIMUM
		A ±1/8	B ±1/32	C ±1/16	D ±1/64	E ±1/8	F ±1/64	
INCH	INCHES	INCHES	INCH	INCHES	INCH	INCHES	INCH	POUNDS
3/4	18	1 1/4	3/4	1 3/16	1/4	3	1/4	50

FIGURE 2 TYPE II PINCH BAR(OFFSET)

Source: https://assist.dla.mil -- Downloaded: 20261021T21:21Z
Check the source to verify that this is the current version before use.



OPTIONAL CONSTRUCTION
(CHAMFER)

SIZE NUMBER	DIMENSIONS									TEST LOAD
	A	B	C	D	E	F	G	H	J	
	+1 -1/2	±1/32	+1/8 -1/16	±1/8	±1/32	MINIMUM	±1/16	MINIMUM	±1/32	MINIMUM
	INCHES	INCH	INCH	INCHES	INCH	INCHES	INCH	INCHES	INCH	POUNDS
1	18	1/16	15/16	1 1/8	5/8	1	5/16	1 1/8	1/16	100
2	24	1/16	1	1 3/8	3/4	1 5/16	7/16	1 3/16	1/16	150
3	36	1/16	1	1 3/8	7/8	1 5/16	7/16	1 3/16	1/16	150

FIGURE 3 TYPE III WRECKING BAR, GOOSENECK, WITH CLAW, AND PINCH POINT

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:21Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Bars: Nonsparking and Nonmagnetic - MIL-B-43407

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:21Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01762

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRE-SS
Natick, Massachusetts 01762

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:21Z
Check the source to verify that this is the current version before use.